

Table 2: Accuracy comparison with respect to the number of iterations K_{init} and K_{on} for fast adaptation on OTB2015.

Method	K_{init}	K_{on}	Succ (%)	Prec (%)	FPS
RT-MDNet	50	15	65.0	88.5	42
	50	5	61.9	83.1	54
	5	15	60.6	80.8	43
	5	5	57.8	77.5	55
MetaRTT	5	5	65.5	89.0	58

Table 3: Analysis of one-shot channel pruning on OTB2015. PR denotes prune rate in this table.

Method	Succ (%)	Prec (%)	FPS	PR (%)
MetaRTT	65.5	89.0	58	0
MetaRTT+Prune	65.1	87.4	65	50 ± 5
- No meta-learning	61.7	83.1	64	49 ± 6
- No adaptive pruning	60.9	81.2	59	53

Meta-learning for one-shot channel pruning Table 1 presents that MetaRTT+Prune is 12% faster than MetaRTT while achieving the almost identical success rate and precision. To analyze the proposed pruning method, we evaluate two variants of pruning strategies. One is the one-shot channel pruning without meta-learning, which optimizes the pruning network using a simulated tracking episode only at the first frame. The other option is the static pruning, which learns a universal channel pruning mask $\mathcal{B}$ offline for every video. Table 3 implies that both the meta-learning with complete simulated episodes and the input adaptive one-shot pruning are essential. It also presents that MetaRTT+Pruning improves speed substantially by removing almost half of the network parameters with minimal accuracy loss.

Comparison with state-of-the-art trackers

We compare variants of our meta-learning algorithm—MetaRTT, MetaRTT+Prune, and MetaRTT+COCO—to recent state-of-the-art algorithms (?: ?; ?; ?; ?; ?). We employ OTB2015 (?) and TempleColor (?) datasets for evaluation. Because our goal is to accelerate tracking algorithms while maintaining their accuracy, the comparison with real-time trackers is more important. So, we highlight real-time trackers with solid lines in the graphs visualizing quantitative performance while the slow methods are represented with dashed lines.

Figure 2 shows that both MetaRTT and MetaRTT+COCO outperform the compared state-of-the-art real-time trackers on OTB2015 in terms of accuracy while MetaRTT+Prune achieves competitive accuracy and improved speed compared to MetaRTT. Figure 3 shows that our trackers present outstanding performance on TempleColor as well. These results imply that the hyperparameters meta-learned on ImageNet-Vid performs well on the standard benchmarks without manual tuning for the benchmarks; we fix all the hyperparameters except the meta-parameters in both datasets. The high accuracy by MetaRTT+COCO also suggests that our tracker can further improve by pretraining with more

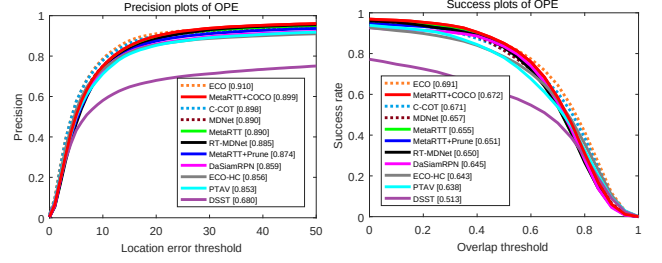


Figure 2: Quantitative results on OTB2015.

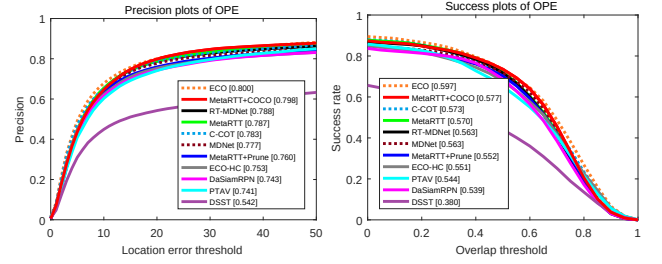


Figure 3: Quantitative results on TempleColor.

Table 4: Results of our algorithms on VOT2016, UAV123.

Method	UAV123 Succ (%)	UAV123 Prec (%)	VOT2016 EAO
RT-MDNet	52.8	77.2	0.322
MetaRTT	56.9	80.9	0.346
MetaRTT+COCO	56.6	80.9	0.350
MetaRTT+Prune	53.7	77.9	0.314

datasets.

Table 4 presents that the variants of our algorithm make significant improvement on other benchmark datasets, VOT2016 (?) and UAV123 (?).

6 Conclusion

We have presented a novel meta-learning framework for object tracking, which is more comprehensive and principled than the technique introduced in previous studies. The proposed meta-learning approach allows to learn fast adaptation and one-shot channel pruning algorithm, which leads to competitive accuracy and substantial speed-up. Our tracker achieves the state-of-the-art performance compared to the existing real-time tracking methods.

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